

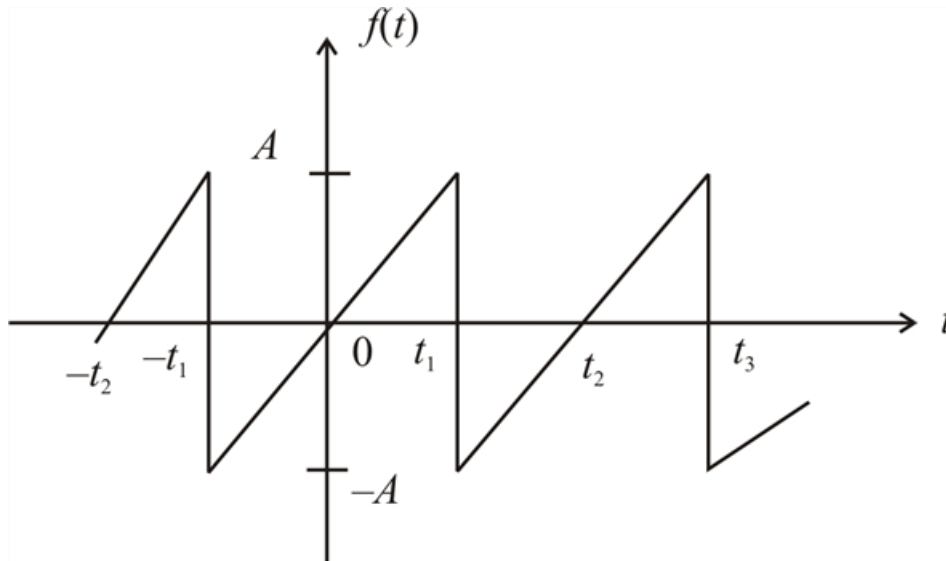
Problem

Design a problem to help other students better understand the amplitude and phase spectra of a given Fourier series.

Step-by-step solution

Step 1 of 5

Consider a periodic signal



Step 2 of 5

Thus, the function is

$$f(t) = \frac{A}{t_1} t \quad -t_1 < t < t_1$$

Since the period is T

$$\omega_0 = \frac{2\pi}{T}$$

The exponential Fourier series coefficients are,

$$C_0 = \frac{1}{T} \int_0^T f(t) dt$$

$$C_0 = \frac{1}{T} \int_{-t_1}^{t_1} \frac{At}{t_1} dt$$

$$C_0 = \frac{A}{t_1 T} \left[\frac{t^2}{2} \right]_{-t_1}^{t_1}$$

$$C_0 = \frac{A}{2t_1 T} [t_1^2 - (-t_1)^2]$$

$$\therefore C_0 = 0$$

Step 3 of 5

$$C_n = \frac{1}{T} \int_0^T f(t) e^{-jn\omega_0 t} dt$$

$$C_n = \frac{1}{T} \int_{-t_1}^{t_1} \frac{At}{t_1} e^{-jn\omega_0 t} dt$$

$$C_n = \frac{A}{t_1 T} \int_{-t_1}^{t_1} t e^{-jn\omega_0 t} dt$$

$$C_n = \frac{A}{t_1 T} \left[\frac{t e^{-jn\omega_0 t}}{-jn\omega_0} \right]_{-t_1}^{t_1} - \int_{-t_1}^{t_1} \frac{e^{-jn\omega_0 t}}{-jn\omega_0} dt$$

$$C_n = \frac{A}{t_1 T} \left[\frac{t_1 e^{-jn\omega_0 t_1} - (-t_1) e^{jn\omega_0 t_1}}{-jn\omega_0} - \frac{e^{-jn\omega_0 t}}{(-jn\omega_0)^2} \right]_{-t_1}^{t_1}$$

$$C_n = \frac{A}{t_1 T} \left[\frac{t_1 (e^{-jn\omega_0 t_1} + e^{jn\omega_0 t_1})}{-jn\omega_0} - \frac{e^{-jn\omega_0 t_1} - e^{jn\omega_0 t_1}}{(-jn\omega_0)^2} \right]$$

$$C_n = \frac{A}{t_1 T} \left[\frac{2jt_1 \cos(n\omega_0 t_1)}{n\omega_0} + \frac{2j \sin(n\omega_0 t_1)}{n^2 \omega_0^2} \right]$$

$$\therefore C_n = \frac{2jA}{t_1 T} \left[\frac{t_1 \cos(n\omega_0 t_1)}{n\omega_0} + \frac{\sin(n\omega_0 t_1)}{n^2 \omega_0^2} \right]$$

Step 4 of 5

Therefore, the exponential Fourier series

$$f(t) = \sum_{\substack{n=-\infty \\ n \neq 0}}^{\infty} \frac{2jA}{t_1 T} \left[\frac{t_1 \cos(n\omega_0 t)}{n\omega_0} + \frac{\sin(n\omega_0 t)}{n^2 \omega_0^2} \right] e^{jn\pi t}$$

The complex frequency spectrum of $f(t)$

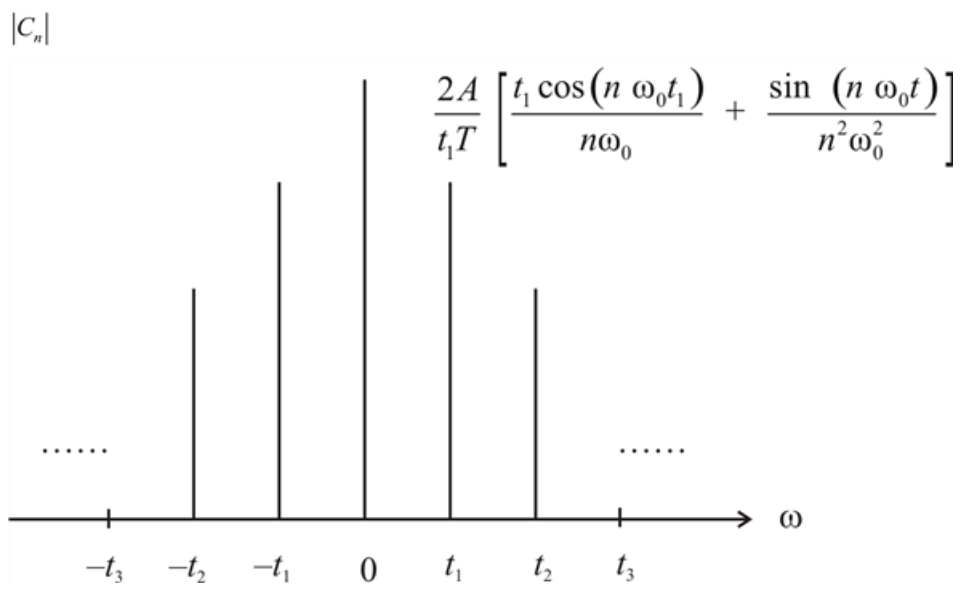
$$C_n = |C_n| \angle \theta_n$$

$$|C_n| = \frac{2A}{t_1 T} \sqrt{\left[\frac{t_1 \cos(n\omega_0 t_1)}{n\omega_0} + \frac{\sin(n\omega_0 t_1)}{n^2 \omega_0^2} \right]^2}$$

$$\theta_n = \angle 90^\circ$$

Step 5 of 5

Amplitude spectrum



Phase spectrum

